



HOW TO USE THIS SET

20 questions, 5 options each (A-E), mixed difficulty, 14 case-based. Attempt every question before checking the answer key on the last page.

● 4 Easy

● 10 Medium

● 6 Hard

1.

EASY

Which test is the more specific marker of liver injury?

- A. AST
- B. ALT
- C. ALP
- D. GGT
- E. Albumin

2.

EASY

Which of these results is outside its normal range?

- A. ALT 40 IU/L
- B. AST 44 IU/L
- C. ALP 110 IU/L
- D. GGT 48 IU/L
- E. Total bilirubin 2.4 mg/dL

3.

EASY

Prothrombin time (PT) evaluates which part of coagulation?

- A. The intrinsic pathway
- B. The extrinsic pathway
- C. Platelet aggregation
- D. Clot breakdown
- E. Bleeding time

4.

EASY

Bilirubin is the yellowish pigment seen in jaundice. It is a byproduct of:

- A. Red blood cell breakdown
- B. Glutathione recycling
- C. Albumin synthesis
- D. Bile salt production
- E. Glycogen storage

5.

MEDIUM

CASE

A 52-year-old man who drinks heavily has AST 160 IU/L and ALT 70 IU/L. ALP and bilirubin are normal.

Which condition does this AST : ALT pattern suggest?

- A. Viral hepatitis
- B. Non-alcoholic liver disease
- C. Gallstone obstruction
- D. Alcoholic liver disease
- E. Bone disease

6.

MEDIUM

CASE

A 67-year-old woman has several hours of very low blood pressure after major bleeding. Next day her ALT is 2,400 IU/L.

Which lecture-listed cause of a very high ALT fits best?

- A. Alcoholic liver disease
- B. Multiple myeloma
- C. Ischemic hepatitis
- D. Vitamin K deficiency
- E. Bone disease

7.

MEDIUM

CASE

A 60-year-old man has crushing chest pain. AST is 130 IU/L. ALT, ALP and bilirubin are normal.

What best explains the isolated rise in AST?

- A. AST is also found in the heart and rises in MI
- B. AST is the most liver-specific of the enzymes
- C. AST is made in the placenta and bile ducts
- D. AST rises whenever bilirubin is about to rise
- E. AST is a clotting factor used up in infarction

8.

MEDIUM

CASE

A healthy woman in late pregnancy has ALP 240 IU/L. ALT, AST, GGT and bilirubin are normal.

Which source of ALP named in the lecture best explains this?

- A. Heart
- B. Kidney
- C. Brain
- D. Skeletal muscle
- E. Placenta

9.

MEDIUM

A man with chronic alcohol consumption has a raised GGT. Which reaction does this enzyme catalyse?

- A. Transfer of an amino group from alanine to α -ketoglutarate
- B. Hydrolysis of phosphate groups at an alkaline pH
- C. Transfer of the γ -glutamyl group from glutathione to an amino acid
- D. Transfer of an amino group from aspartate to α -ketoglutarate
- E. Conversion of unconjugated bilirubin to conjugated bilirubin

10.

MEDIUM

CASE

A 58-year-old woman with cirrhosis has a low serum albumin and a prolonged PT.

Both results reflect failure of which liver function listed in the lecture?

- A. Digestion
- B. Detoxification
- C. Storage
- D. Synthesis
- E. Immune function

11.

MEDIUM

CASE

A 20-year-old man has yellow eyes. Total bilirubin is 4.8 mg/dL and conjugated (direct) bilirubin is 0.3 mg/dL. Liver enzymes are normal.

Which fraction is raised, and how does it travel in the blood?

- A. Unconjugated (indirect), as a bilirubin–albumin complex
- B. Conjugated (direct), as a bilirubin–albumin complex
- C. Unconjugated (indirect), bound to γ -globulins
- D. Conjugated (direct), bound to glutathione
- E. Both fractions equally, dissolved freely in plasma

12.

MEDIUM

CASE

A 70-year-old man takes an anticoagulant tablet for an irregular heartbeat. His INR is 2.8. ALT, AST, ALP, bilirubin and albumin are normal.

Which lecture-listed cause of a prolonged PT/INR is most likely?

- A. Cirrhosis
- B. Acute liver failure
- C. Warfarin therapy
- D. Hepatitis
- E. Multiple myeloma

13.

MEDIUM

Two laboratories use different reagents and report different PT values, in seconds, for the same blood sample. Which measure makes their results comparable?

- A. Total protein
- B. INR
- C. Conjugated bilirubin
- D. AST : ALT ratio
- E. GGT

14.

MEDIUM

CASE

A 68-year-old man has a serum globulin of 6.0 g/dL with raised γ -globulins. Albumin, bilirubin and liver enzymes are normal, and he has no infection.

Which lecture-listed cause is most likely?

- A. Cirrhosis
- B. Chronic hepatitis
- C. Gallstone obstruction
- D. Multiple myeloma
- E. Vitamin K deficiency

15.

HARD

CASE

A laboratory assay supplies alanine and α -ketoglutarate, then measures how fast the amino group is transferred in a patient's serum.

Which enzyme is being measured, and what does the lecture say about it?

- A. AST — found mostly in the liver, heart and muscle
- B. ALP — a non-specific marker of liver disease
- C. GGT — important in glutathione recycling
- D. AST — more liver-specific than ALT
- E. ALT — more liver-specific than AST

16.

HARD CASE

Two patients have a prolonged PT. Patient 1: albumin 2.5 g/dL with raised γ -globulins. Patient 2: albumin 4.4 g/dL, normal globulin, normal liver enzymes, a very poor diet and no medication.

Which pairing of causes fits best?

- A. 1 = vitamin K deficiency; 2 = chronic liver disease
- B. 1 = chronic liver disease; 2 = vitamin K deficiency
- C. 1 = warfarin therapy; 2 = acute liver failure
- D. 1 = multiple myeloma; 2 = cirrhosis
- E. 1 = chronic liver disease; 2 = warfarin therapy

17.

HARD CASE

A 62-year-old man has an ALP of 450 IU/L on routine testing. His doctor wants to know whether it comes from the bile ducts or from bone.

Which additional result most favours a bile-duct source?

- A. A raised GGT
- B. A normal total bilirubin
- C. An INR of 1.0
- D. An albumin of 4.2 g/dL
- E. A normal globulin

18.

HARD CASE

Five patients have these results, written as AST / ALT / ALP in IU/L, then total bilirubin in mg/dL.

Which patient best fits cholestasis (obstructive jaundice)?

- A. 22 / 25 / 90; bilirubin 0.7
- B. 30 / 28 / 100; bilirubin 5.2
- C. 85 / 95 / 540; bilirubin 6.5
- D. 210 / 80 / 110; bilirubin 1.0
- E. 38 / 40 / 115; bilirubin 1.1

19.

HARD CASE

A boy has an inherited deficiency of a single clotting factor, yet his PT is normal.

Going by the lecture's list of factors that PT depends on, which factor is he most likely to lack?

- A. Factor II
- B. Factor V
- C. Factor VII
- D. Factor X
- E. Factor VIII

20.

HARD CASE

A laboratory report shows total protein 7.8 g/dL and albumin 2.6 g/dL. No other protein fractions are given.

Which interpretation is correct?

- A. Albumin and globulin are both normal
- B. Albumin is low; globulin is normal
- C. Albumin is normal; globulin is high
- D. Albumin is low; globulin is high
- E. Albumin is high; globulin is low

ANSWER KEY & RATIONALE

Don't peek until you've committed to an answer on every question.

Q	Ans	Why
1	B	ALT — ALT is found mainly in the liver and is more specific for liver injury than AST.
2	E	Total bilirubin 2.4 mg/dL — Total bilirubin is normally 0.2–1.2 mg/dL. The others sit inside 5–45 (ALT, AST), 30–120 (ALP) and 5–50 IU/L (GGT).
3	B	The extrinsic pathway — PT evaluates the extrinsic pathway, which depends on clotting factors II, V, VII and X.
4	A	Red blood cell breakdown — Bilirubin is a byproduct of red blood cell breakdown.
5	D	Alcoholic liver disease — AST greater than ALT points to alcoholic liver disease; ALT greater than AST points to viral or non-alcoholic disease.
6	C	Ischemic hepatitis — Very high ALT is most often caused by viral hepatitis, ischemic hepatitis, or injury from drugs or other chemicals. Loss of blood supply fits the ischemic cause.
7	A	AST is also found in the heart and rises in MI — AST is less liver-specific: it is also in heart, kidneys, skeletal muscle and brain, and MI is a listed cause of a raised level. The normal ALT argues against liver injury.
8	E	Placenta — ALP is found mainly in the liver, bones, bile ducts and placenta. The other four organs are sources of AST or GGT, not ALP.
9	C	Transfer of the γ-glutamyl group from glutathione to an amino acid — GGT transfers the γ -glutamyl group from glutathione to an amino acid. The amino-group transfers belong to ALT and AST; phosphate hydrolysis belongs to ALP.
10	D	Synthesis — The liver produces essential proteins, including albumin, and the clotting factors that PT depends on.
11	A	Unconjugated (indirect), as a bilirubin-albumin complex — Conjugated 0.3 is normal (0.0–0.4). The remaining 4.5 mg/dL is unconjugated (normal 0.2–0.8), which the lecture describes as a bilirubin-albumin complex.
12	C	Warfarin therapy — Warfarin therapy prolongs PT/INR. Every other liver test is normal, so a liver cause is unlikely.
13	B	INR — INR is a standardised way of reporting PT so that results are comparable across labs. Normal is about 1.0.
14	D	Multiple myeloma — High γ -globulins occur in chronic liver disease, infection and multiple myeloma. A normal albumin and normal enzymes argue against chronic liver disease, and infection is excluded.
15	E	ALT — more liver-specific than AST — ALT transfers the amino group from alanine to α -ketoglutarate, and is more specific for liver injury than AST. AST uses aspartate.
16	B	1 = chronic liver disease; 2 = vitamin K deficiency — Low albumin with high γ -globulins is the chronic liver disease pattern. Patient 2 has a normal liver panel and takes no warfarin, leaving vitamin K deficiency.
17	A	A raised GGT — ALP is non-specific (liver, bones, bile ducts, placenta). GGT is made in the liver, kidney and biliary tract and rises in liver and bile duct disorders — bone is not one of its sources.
18	C	85 / 95 / 540; bilirubin 6.5 — This mirrors the lecture's case 3: a very high ALP with raised bilirubin and only modestly raised AST and ALT. A is normal, B is a bilirubin rise alone, D is the AST > ALT pattern.
19	E	Factor VIII — PT depends on factors II, V, VII and X. Factor VIII is not on that list, so its deficiency would leave PT normal.
20	D	Albumin is low; globulin is high — Total protein = albumin + globulins, so globulin is $7.8 - 2.6 = 5.2$ g/dL (normal 2.5–3.5). Low albumin with high globulin is the chronic liver disease pattern.